

# Pharma Agentic AI – Analysis Report

Automated opportunity assessment powered by Agentic AI (Master + Worker agents).

|                  |                                                                                                                  |              |                                              |
|------------------|------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------|
| Query            | Evaluate MOL2 in the Indian Chronic Kidney Disease space, highlighting decline risk, unmet needs and whitespace. |              |                                              |
| Molecule         | MOL2                                                                                                             | Therapy area | Chronic Kidney Disease                       |
| Region / Country | India                                                                                                            | Agents run   | MARKET EXIM<br>PATENT TRIALS<br>INTERNAL WEB |
| Generated on     | 13 Dec 2025, 10:53 PM                                                                                            |              |                                              |

## Executive Summary

- Market & growth view: Total market size: 10589.9 Mn USD with average 5Y CAGR 6.2%. Records: 19.
- Supply and trade dynamics: Total imports: 24269.6 tonnes, total exports: 26074.8 tonnes across 260 records.
- IP barriers and freedom-to-operate: Found 156 relevant patents. Earliest expiry year: 2003.
- Clinical development and pipeline: Found 16 clinical trials matching the criteria.
- Internal strategy and unmet needs: Evaluating MOL2 in Chronic Kidney Disease space highlighting unmet needs, repurposing opportunities and risks in the Indian market.
- Guidelines and real-world evidence: The 2024 CHRONIC guideline recommends step-wise treatment for Chronic Kidney Disease (CKD) based on disease severity and patient phenotype. Combination therapy is advised when monotherapy fails to achieve control. Real-world evidence highlights adherence challenges, access barriers, and regional disparities in CKD outcomes. Patient-reported outcomes emphasize unmet needs in symptom control and quality of life.

## Agent-level Insights

|                      |
|----------------------|
| IQVIA Insights Agent |
| High-level summary   |

Total market size: 10589.9 Mn USD with average 5Y CAGR 6.2%. Records: 19.

Structured details

Table:

- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2019 **sales\_usd\_mn:** 316.75 **cagr\_5y:** -4.85 **competitors:** 22 **patient\_burden\_index:** 0.26
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2024 **sales\_usd\_mn:** 861.42 **cagr\_5y:** 15.4 **competitors:** 16 **patient\_burden\_index:** 0.68
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2022 **sales\_usd\_mn:** 976.84 **cagr\_5y:** 1.24 **competitors:** 22 **patient\_burden\_index:** 0.97
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2021 **sales\_usd\_mn:** 497.54 **cagr\_5y:** 8.97 **competitors:** 14 **patient\_burden\_index:** 0.95
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2017 **sales\_usd\_mn:** 480.25 **cagr\_5y:** 8.41 **competitors:** 22 **patient\_burden\_index:** 0.73
- ... (5 more records)

EXIM Trade Agent

High-level summary

Total imports: 24269.6 tonnes, total exports: 26074.8 tonnes across 260 records.

Structured details

Table:

- **molecule:** MOL2 **formulation:** Tablet **country:** India **year:** 2015 **import\_tonnes:** 154.23 **export\_tonnes:** 104.66
- **molecule:** MOL2 **formulation:** Patch **country:** India **year:** 2015 **import\_tonnes:** 64.44 **export\_tonnes:** 184.9

- **molecule:** MOL2 **formulation:** Capsule **country:** India **year:** 2019 **import\_tonnes:** 19.83 **export\_tonnes:** 81.0
- **molecule:** MOL2 **formulation:** Ointment **country:** India **year:** 2021 **import\_tonnes:** 2.47 **export\_tonnes:** 121.37
- **molecule:** MOL2 **formulation:** API **country:** India **year:** 2017 **import\_tonnes:** 110.77 **export\_tonnes:** 87.92
- ... (5 more records)

## Patent Landscape Agent

### High-level summary

Found 156 relevant patents. Earliest expiry year: 2003.

### Structured details

#### Patents:

- **patent\_id:** US1000426 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** GlobalBio **filing\_year:** 2011 **expiry\_year:** 2031 **status:** Granted **fto\_flag:** Medium
- **patent\_id:** US1001908 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** ABC Pharma **filing\_year:** 2005 **expiry\_year:** 2021 **status:** Pending **fto\_flag:** High
- **patent\_id:** US1003136 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** NovaCure **filing\_year:** 2005 **expiry\_year:** 2027 **status:** Pending **fto\_flag:** Medium
- **patent\_id:** US1003185 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** ABC Pharma **filing\_year:** 2018 **expiry\_year:** 2031 **status:** Expired **fto\_flag:** Low
- **patent\_id:** US1003444 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** ABC Pharma **filing\_year:** 2017 **expiry\_year:** 2029 **status:** Expired **fto\_flag:** Low
- ... (5 more records)

## Clinical Trials Agent

High-level summary

Found 16 clinical trials matching the criteria.

Structured details

Table:

- **trial\_id:** CT-005276 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase I **status:** Recruiting **sponsor:** ABC Pharma **country:** India
- **trial\_id:** CT-018940 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase I **status:** Suspended **sponsor:** ABC Pharma **country:** India
- **trial\_id:** CT-019942 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase II **status:** Active **sponsor:** PrimeCare **country:** India
- **trial\_id:** CT-022598 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase I **status:** Completed **sponsor:** PrimeCare **country:** India
- **trial\_id:** CT-030017 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase IV **status:** Terminated **sponsor:** XYZ Pharma **country:** India
- ... (5 more records)

Internal Knowledge Agent

High-level summary

Evaluating MOL2 in Chronic Kidney Disease space highlighting unmet needs, repurposing opportunities and risks in the Indian market.

Structured details

Documents:

- asthma\_strategy\_2024.pdf

- respiratory\_field\_insights.pdf

**Key Opportunities:**

- Potential repurposing as a once-daily oral formulation for better adherence in CKD patients
- Addressing pain points in CKD patients with cardiovascular comorbidities
- Targeting proteinuria and associated complications as a novel therapeutic strategy
- Low-dose options for pediatric CKD patients to manage chronic conditions
- Steroid-sparing therapy to mitigate side effects in CKD patients

**Key Risks:**

- Limited evidence on long-term safety of MOL2 in CKD patients
- Potential interaction with nephrotoxic medications used in CKD management
- Increased cardiac risk in CKD patients with concomitant cardiovascular conditions
- Regulatory scrutiny due to potential misuse in non-CKD indications
- Operational constraints in managing complex CKD patient populations, including monitoring and comorbidity management

**Web Intelligence Agent**

**High-level summary**

The 2024 CHRONIC guideline recommends step-wise treatment for Chronic Kidney Disease (CKD) based on disease severity and patient phenotype. Combination therapy is advised when monotherapy fails to achieve control. Real-world evidence highlights adherence challenges, access barriers, and regional disparities in CKD outcomes. Patient-reported outcomes emphasize unmet needs in symptom control and quality of life.

**Structured details**

**Links:**

- **title:** Chronic Kidney Disease guideline (synthetic) **url:** <https://example-guideline.com>

- **title:** Chronic Kidney Disease RWE summary **url:** <https://example-rwe.com>

#### Key Updates:

- Combination therapy is recommended for CKD treatment when monotherapy fails

- Non-pharmacological measures are core components of CKD management, including lifestyle, diet, and exercise

- Suboptimal adherence and regional disparities affect CKD outcomes

- Patient-reported outcomes highlight unmet needs in symptom control and quality of life

- Access, affordability, and health literacy remain key barriers for CKD patients

This report is generated by the Pharma Agentic AI prototype for EY Hackathon 6.0. Insights are based on simulated / mock data sources and should not be used as a substitute for validated market research.